

# Scale generation: independent batch workers

First production submission: September 9, 21:32 UTC. The 18-hour audit completed while generation continued.

Frozen plan: 4.82 million raw backbones at lengths 60–500, four class arms, 768 independently admitted worker queues. East02 has 256 H100s. RNO2A transfer approval remains pending.

Preserve all backbones and sequences, including rejects. Quality-pass documents require decontamination and global diversity selection before training use.

# Exact scale recipe

Compiled ProteinA 200M no-triangle checkpoints: short through 250 aa, long from 251 aa. Every integer length 60–500; unconditional, alpha, beta and mixed class requests; 400 steps; one CA ProteinMPNN sequence per backbone, temperature 0.1.

ESMFold refolds each sequence (four recycles). Require CA self-consistency  $\text{RMSD} \leq 2 \text{ \AA}$ ,  $\text{pLDDT} \geq 70$  and cis-proline-aware CA geometry checks. Contacts-v1 labels use refolded geometry. Each independent single-GPU worker alternates sampling and refolding by case.

# Scale results — September 14, 19:19 UTC

2,887,824 saved backbones (59.9% of the frozen manifest); 2,817,968 saved sequences; 2,816,848 committed refolds; 1,717,433 quality-pass provisional documents (61.0% of refolds).

136 H100 pods active and 377 waiting, all batch priority; 255 of 768 root jobs succeeded and none terminally failed. Measured Iris task-attempt time is 6,724.9 H100-hours including setup and retries.

The 3,008-candidate audit estimates 23.86% quality/reference retention before global clustering. Its fine-cluster cap retained 838/858 eligible structures, projecting 656k retained documents so far and 1.12M at manifest completion if sample behavior holds. Full-corpus clustering remains required.

The random 64-structure montage spans 65–500 aa; median pLDDT 89.6 and self-consistency RMSD 1.08 Å. Visual variety includes compact alpha, beta-rich and mixed structures, alongside open or elongated quality passes that merit compactness analysis.

Sources: RUN\_SUMMARY.md; data/scale-20260909/reports/snapshot-20260914T191922Z.json; data/scale-20260909/review-18h/; and data/scale-20260909/structure-montage-sample.csv.

# Proteina initial screen

We filed experiment #278 and completed a 1,536-backbone screen on Iris H100s. The end-to-end pipeline produces valid contacts-v1 documents, and class conditioning changes secondary-structure composition.

494 documents survive quality, evaluation decontamination and fine-cluster capping. The user authorized a larger run to measure retention and diversity. This is an initial-screen result, not completion of all proposed pilot arms.

# What survived

Proteina CA traces → CA ProteinMPNN → ESMFold full backbone → quality checks → evaluation  
decontamination → structural selection → contacts-v1.

1,536 raw; 1,044 quality-pass; 523 decontaminated; 494 selected in 453 fine clusters.  
Retention at 60/100/200/300/400/500 aa is 53% / 53% / 34% / 21% / 21% / 11%.

All selected sequences, contact counts and endpoints validate. No documents are truncated.  
No generated data entered training.

# Diversity: composition control, unresolved coverage benefit

Conditioning changes alpha/beta composition, but this is not independent CATH topology assignment. Fine-cluster effective count is  $0.965\times$  a matched unconditional sample (108 per arm; subsampling range 0.881–1.057).

The  $1.5\times$  target is poorly calibrated here: the control already has about 95 effective clusters out of 108 samples, leaving a maximum median gain of only  $1.14\times$ . This screen cannot rule out a benefit from larger samples or A/T conditioning.

Next design: diversity accumulation curves and architecture/topology coverage against a frozen natural-data baseline, plus long-chain rejection analysis.

# Measured cost replaces the initial estimate

The screen and auxiliary probes consumed 4.706 H100-hours including setup and failed starts. All GPU jobs finished.

At screening yield, one million retained documents uniformly spanning 60–500 aa project to 13,382 H100-hours (17.4 days on 32 GPUs), including 20% overhead. Balancing accepted contributions across all four requested arms projects to 15,823 hours.

Illustrative 2–4/H100 – *houraccounting* gives 26,800–53,500 for the primary projection. Fleet is prepaid; these are not contract quotes. Million-scale duplicate growth, CPU selection and storage/transfer costs remain unmeasured.

# Auxiliary probes and scope

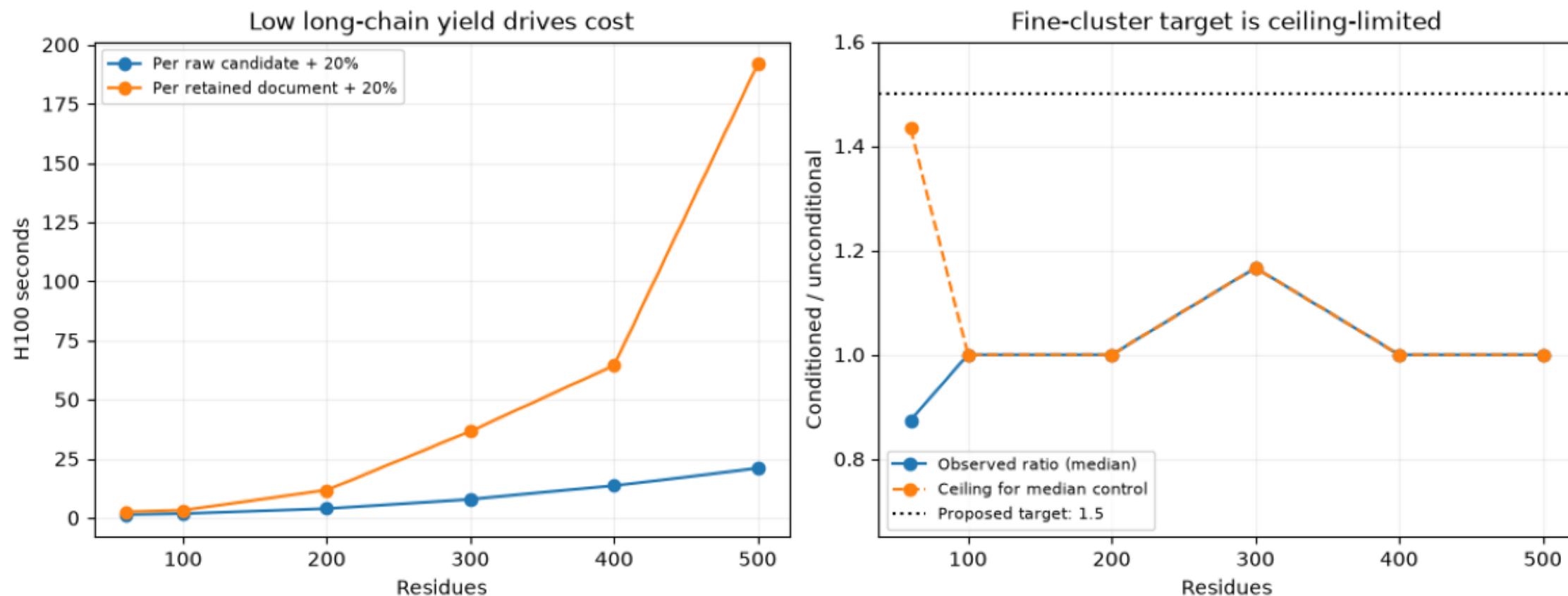
500-aa TF32 sampling: 5.00 vs 10.16 seconds/backbone; 30/48 vs 27/48 pass quality. Matched original seeds can diverge, and MPNN seeds differ between arms: promising speed, not established quality noninferiority.

A second sequence attempt rescued 5 of 32 identical 500-aa backbones: first 18/32; best-of-two 23/32; 231 extra inference seconds before decontamination.

A/T labels, higher noise, the 400M triangle model and a larger pilot remain untested; 250/251-aa crossover canaries subsequently passed integration checks. A scale run is now authorized; issue #278 and draft PR #282 retain the code, timing records and small result artifacts.

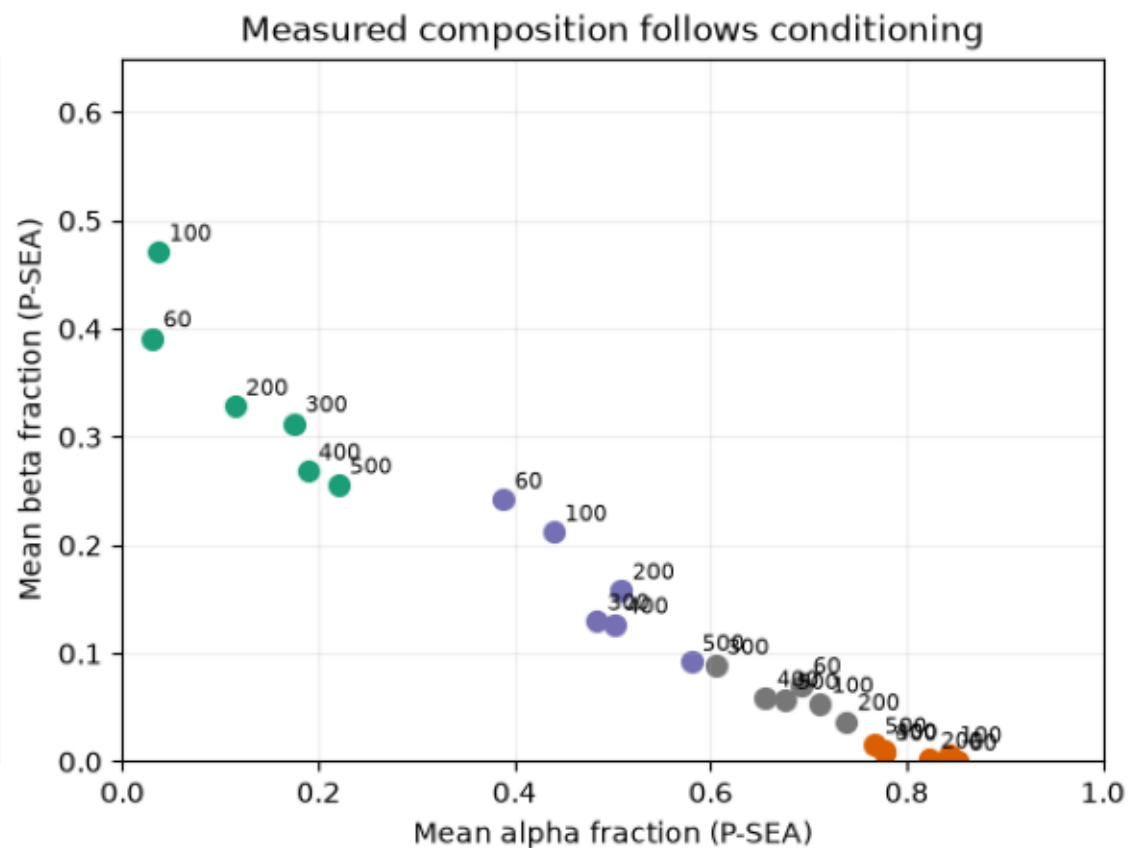
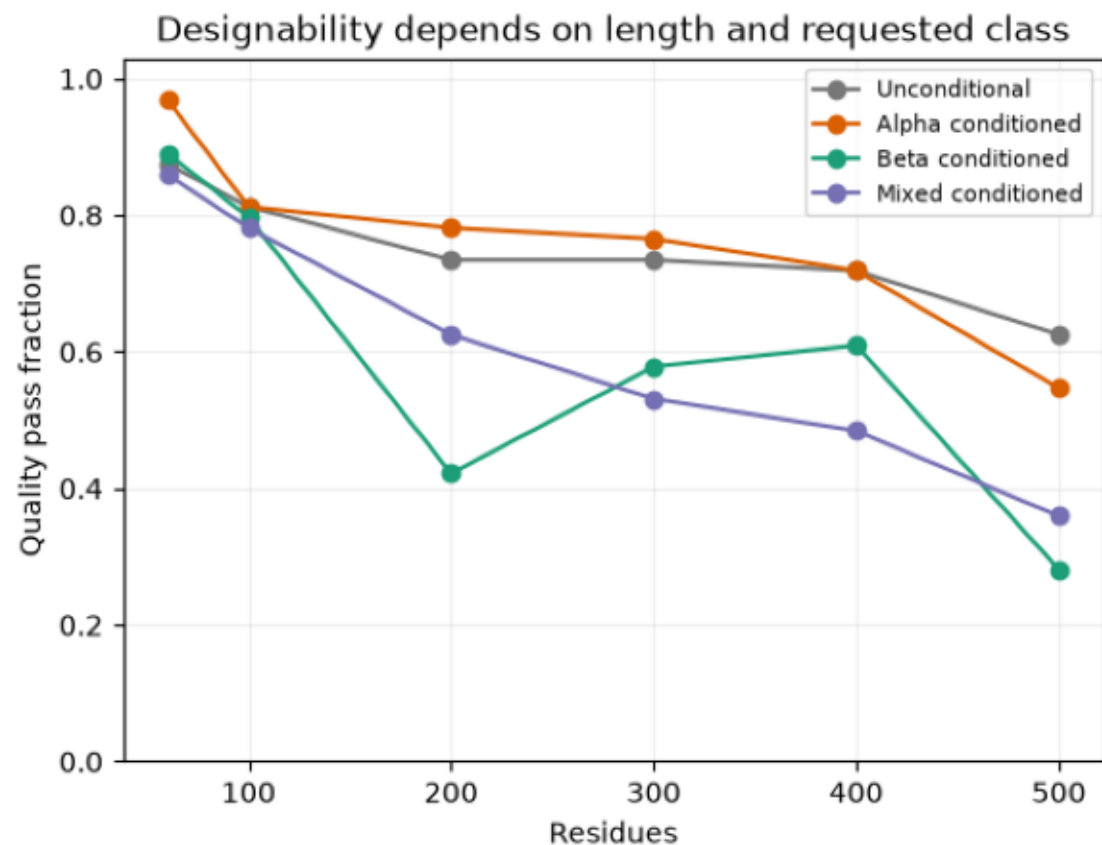
## cost-and-diversity.png

Reference-precision stage costs and final retained yield. Diversity uses equal-size length-matched samples after decontamination. The ceiling assumes every conditioned sample occupies a unique cluster; this exposes limited power of the proposed small-screen target.



# quality-and-composition.png

One ProteinMPNN/ESMFold attempt per backbone; final cis-aware quality checks. Composition averages all refolded candidates and is an independent P-SEA measurement, not a CATH assignment. Labels on the right denote sequence length.



Fixed-seed simple random sample of 64 quality-passing ESMFold refolds from the production audit pool. PyMOL cartoons are independently oriented and scaled and colored from N terminus (blue) to C terminus (red). Panel labels report length, requested conditioning arm, mean pLDDT, and full-chain C-alpha self-consistency RMSD to the generated backbone.

64 random quality-passing ProteinA refolds



```
$ uv run --with pymol-open-source==3.2.0a0 python plot_structure_montage.py --count 64 --seed 2786401
```

## retention.png

Screening-set retention. Duplicate growth at 100k/1M remains unmeasured; these are not production-yield forecasts.

